

- LocalStack enables faster deploy-test-redeploy cycles, allowing for rapid iteration and development.
- It provides easier debugging capabilities and facilitates the replication of bugs locally, leading to improved software quality.
- You no longer need to create individual AWS developer accounts for each user, streamlining administrative tasks and reducing complexity.
- You can overcome barriers in regulated environments where access to the cloud is often restricted, enabling developers to leverage its benefits regardless of limitations.
- Finally, cost considerations play a significant role. You don't need to worry about uncleaned resources since they are created locally and can be easily cleaned.

To install LocalStack on your machine, you need Python (and *pip* package manager) and Docker. Let us install LocalStack using *pip*:

```
$ pip3 install localstack
```

The above command will install the *localstack-cli* which runs the Docker image that packages the LocalStack runtime. Start LocalStack in a detached mode by running the following command:

```
$ localstack start -d
```

It will initialise LocalStack in a detached mode, and now you have a full suite of local AWS services that you can use for testing and playing around.

You can take LocalStack for a run by installing AWS CLI on your machine. After installing AWS CLI, run the following commands to create an S3 bucket, add some content to your bucket, and check its content:

```
$ aws --endpoint-url=http://localhost:4566 s3 mb s3://test
make_bucket: test
```

```
$ echo "hello world" > /tmp/hello-world
```

```
$ aws --endpoint-url=http://localhost:4566 s3 cp /tmp/hello-
world s3://test/hello-world
upload: ../../tmp/hello-world to s3://test/hello-world
```

```
$ aws --endpoint-url=http://localhost:4566 s3 ls s3://test/
2023-05-27 15:39:38      12 hello-world
```

If you notice closely, we have configured the *--endpoint-url* flag to point towards *http://localhost:4566*. This is the edge URL where all LocalStack services are available. You can point your AWS CLI or any third-party integration towards this URL, and it will route your AWS API requests to LocalStack instead.

Building on LocalStack — S3 notifications with SQS

We will create a simple S3 and SQS integration to set up a notification configuration using AWS CLI. We will use S3 Event Notifications to send event notification messages to an SQS queue and test the integration locally with LocalStack, without creating any real cloud resource or setting up an AWS account. We assume you are operating an OSX or a Linux distribution for this tutorial. If you are using Windows, using Windows Subsystem for Linux (WSL) is preferable.

Start LocalStack using the *localstack* CLI. Let us specify a few configurations for our S3 Bucket, SQS queue, and AWS region:

```
$ export BUCKET_NAME=s3-event-notification
$ export AWS_REGION=us-east-1
$ export SQS_QUEUE=queue-notification
```

Next, create an SQS queue using AWS CLI:

```
$ aws --endpoint-url=http://localhost:4566 sqs create-queue
--queue-name $SQS_QUEUE
{
  "QueueUrl": "http://localhost:4566/000000000000/queue-
notification"
}
```

You can fetch the Queue ARN (Amazon Resource Name) to specify the value while configuring the bucket notification:

```
$ export SQS_QUEUE_ARN=$(aws --endpoint-url=http://
localhost:4566 sqs get-queue-attributes\
--attribute-name QueueArn --queue-
url=http://localhost:4566/000000000000/"$SQS_QUEUE"\
| sed 's/"QueueArn"/\n"QueueArn"/g' | grep
"QueueArn" | awk -F "'QueueArn':" '{print $2}' | tr -d "'"
| xargs)
$ echo $SQS_QUEUE_ARN
arn:aws:sqs:us-east-1:000000000000:queue-notification
```

Let us create an S3 bucket now and apply the notification configuration:

```
$ aws --endpoint-url=http://localhost:4566 s3api create-
bucket --bucket $BUCKET_NAME --region $AWS_REGION
{
  "Location": "/s3-event-notification"
}
$ aws --endpoint-url=http://localhost:4566 s3api put-bucket-
notification-configuration \
--bucket $BUCKET_NAME\
--notification-configuration '{
```